LINUX PROJECT

Sean Singleton

R00221306

Table of Contents

1. User Accounts

Create 2 user accounts

With sudo access:
sugbo
password: password

```
seansingleton@ubuntu:~$ sudo adduser ssingleton
[sudo] password for seansingleton:
Adding user `ssingleton' ...
Adding new group `ssingleton' (1001) ...
Adding new user `ssingleton' (1001) with group `ssingleton' ...
Creating home directory `/home/ssingleton' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for ssingleton
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
seansingleton@ubuntu:~$ sudo usermod -aG sudo ssingleton
seansingleton@ubuntu:~$ su ssingleton
Password:
To run a command as administrator (user "root"), use "sudo <command>"
See "man sudo_root" for details.

ssingleton@ubuntu:/home/seansingleton$ exit
exit
```

Created the user using “sudo adduser” then for ssingleton I gave this user sudo access so I used the command “sudo usermod -aG sudo ssingleton”

Without sudo access:
achen
Password:password1

```
seansingleton@ubuntu:~$ sudo adduser achen
Adding user `achen' ...
Adding new group `achen' (1002) ...
Adding new user `achen' (1002) with group `achen' ...
Creating home directory `/home/achen' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for achen
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
```

With this user I just created it with the command "sudo adduser achen" and we don't want this user to have sudo access so we leave it as it is.

2. HDD Configuration

```
ssingleton@ssingleton-virtual-machine:~$ sudo parted -l
[sudo] password for ssingleton:
Model: VMware, VMware Virtual S (scsi)
Disk /dev/sda: 21.5GB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Disk Flags:

Number  Start   End     Size    File system  Name                  Flags
 1      1049kB  2097kB  1049kB                                     bios_grub
 2      2097kB  540MB   538MB   fat32        EFI System Partition  boot, esp
 3      540MB   21.5GB  20.9GB  ext4


Error: /dev/sdb: unrecognised disk label
Model: VMware, VMware Virtual S (scsi)
Disk /dev/sdb: 21.5GB
Sector size (logical/physical): 512B/512B
Partition Table: unknown
Disk Flags:

Error: /dev/sdc: unrecognised disk label
Model: VMware, VMware Virtual S (scsi)
Disk /dev/sdc: 21.5GB
Sector size (logical/physical): 512B/512B
Partition Table: unknown
Disk Flags:

ssingleton@ssingleton-virtual-machine:~$ sudo parted /dev/sdb
GNU Parted 3.4
Using /dev/sdb
Welcome to GNU Parted! Type 'help' to view a list of commands.
(parted) mktable gpt
(parted) mkpart 1 0G 5G
(parted) mkpart 2 5G 10G
(parted) mkpart 3 10G 15G
(parted) mkpart 4 15G 20G
(parted) print
Model: VMware, VMware Virtual S (scsi)
Disk /dev/sdb: 21.5GB
Sector size (logical/physical): 512B/512B
Partition Table: gpt
Disk Flags:

Number  Start   End     Size    File system  Name  Flags
 1      1049kB  5000MB  4999MB               1
 2      5000MB  10.0GB  5001MB               2
 3      10.0GB  15.0GB  5000MB               3
 4      15.0GB  20.0GB  5000MB               4
```

```
ssingleton@ssingleton-virtual-machine:~$ sudo mkfs.ext4 /dev/sdb1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 1220352 4k blocks and 305216 inodes
Filesystem UUID: 48730d40-0efa-42f0-a98c-ad3a8de4ebae
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

ssingleton@ssingleton-virtual-machine:~$ sudo mkfs.ext4 /dev/sdb2
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 1220864 4k blocks and 305216 inodes
Filesystem UUID: e0f4e55e-c4de-4a40-9481-96d8bc6534d6
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

ssingleton@ssingleton-virtual-machine:~$ sudo mkfs.ext4 /dev/sdb3
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 1220608 4k blocks and 305216 inodes
Filesystem UUID: 2b0ddff5-c4ea-499c-92fa-37d2bb12729b
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

ssingleton@ssingleton-virtual-machine:~$ sudo mkfs.ext4 /dev/sdb4
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 1220608 4k blocks and 305216 inodes
Filesystem UUID: 16afdbdc-2f0e-443e-85a6-e09ff2b86203
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
ssingleton@ssingleton-virtual-machine:~$ sudo mkdir /sugbo /achen /shared /backups
```

```
  GNU nano 6.2                                  /etc/fstab
# /etc/fstab: static file system information.
#
# Use 'blkid' to print the universally unique identifier for a
# device; this may be used with UUID= as a more robust way to name devi
# that works even if disks are added and removed. See fstab(5).
#
# <file system> <mount point>   <type>  <options>       <dump>  <pass>
# / was on /dev/sda3 during installation
UUID=67e20807-e272-45bb-a4d0-bd1a809b14b9 /               ext4    error
# /boot/efi was on /dev/sda2 during installation
UUID=9588-B6DB  /boot/efi       vfat    umask=0077      0       1
/swapfile                                 none            swap    sw
/dev/fd0        /media/floppy0  auto    rw,user,noauto,exec,utf8 0
UUID=7951cd5d-ee10-4110-ae6f-9a0a7da4c2da /mnt/sugbo ext4 user 0 0
UUID=e23b0fc6-90aa-45d6-90ab-f34cf1229a78 /mnt/achen ext4 user 0 0
UUID=0544ab8d-8c0a-4970-aca5-1d97d4a00698 /mnt/shared ext4 user 0 0
UUID=66e6e379-6bec-40d8-8a7e-cb0fb6870e17 /mnt/backups ext4 nouser 0 0
```

```
ssingleton@ssingleton-virtual-machine:~$ sudo blkid
[sudo] password for ssingleton:
/dev/sda3: UUID="67e20807-e272-45bb-a4d0-bd1a809b14b9" BLOCK_SIZE="4096" TY
xt4" PARTUUID="28fed0b1-ad69-4d97-83a9-cc939ca27357"
/dev/loop1: TYPE="squashfs"
/dev/loop8: TYPE="squashfs"
/dev/sdb4: UUID="66e6e379-6bec-40d8-8a7e-cb0fb6870e17" BLOCK_SIZE="4096" TY
xt4" PARTLABEL="4" PARTUUID="b91b1e0a-d6b2-417f-91af-42a702e04bf6"
/dev/sdb2: UUID="e23b0fc6-90aa-45d6-90ab-f34cf1229a78" BLOCK_SIZE="4096" TY
xt4" PARTLABEL="2" PARTUUID="856014d7-5b37-4ca7-b860-72d37215c3f8"
/dev/sdb3: UUID="0544ab8d-8c0a-4970-aca5-1d97d4a00698" BLOCK_SIZE="4096" TY
xt4" PARTLABEL="3" PARTUUID="ee42007e-6f57-4b4a-9d59-955ef9d6f460"
/dev/sdb1: UUID="7951cd5d-ee10-4110-ae6f-9a0a7da4c2da" BLOCK_SIZE="4096" TY
xt4" PARTLABEL="1" PARTUUID="caa7a82f-2309-4786-a46a-bf25742bf270"
```

```
ssingleton@ssingleton-virtual-machine:~$ sudo mount -a
```

```
ssingleton@ssingleton-virtual-machine:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           389M  2.2M  387M   1% /run
/dev/sda3        20G   11G  7.3G  60% /
tmpfs           1.9G     0  1.9G   0% /dev/shm
tmpfs           5.0M  4.0K  5.0M   1% /run/lock
/dev/sda2       512M  5.3M  507M   2% /boot/efi
tmpfs           389M  2.4M  387M   1% /run/user/1000
/dev/sr0        127M  127M     0 100% /media/ssingleton/CDROM
/dev/sr1        3.6G  3.6G     0 100% /media/ssingleton/Ubuntu 22.04.1 LTS amd6
4
/dev/sdb1       4.6G   24K  4.3G   1% /mnt/sugbo
/dev/sdb2       4.6G   24K  4.3G   1% /mnt/achen
/dev/sdb3       4.6G   24K  4.3G   1% /mnt/shared
/dev/sdb4       4.6G   24K  4.3G   1% /mnt/backups
```

Permissions:

```
ssingleton@ssingleton-virtual-machine:/mnt$ sudo chmod 1777 shared
[sudo] password for ssingleton:
ssingleton@ssingleton-virtual-machine:/mnt$ ls -l
total 16
drwxr-xr-x 3 root root 4096 Dec  6 22:15 achen
drwxr-xr-x 3 root root 4096 Dec  6 22:15 backups
drwxrwxrwt 3 root root 4096 Dec  6 22:15 shared
drwxr-xr-x 3 root root 4096 Dec  6 22:15 sugbo
ssingleton@ssingleton-virtual-machine:/mnt$
```

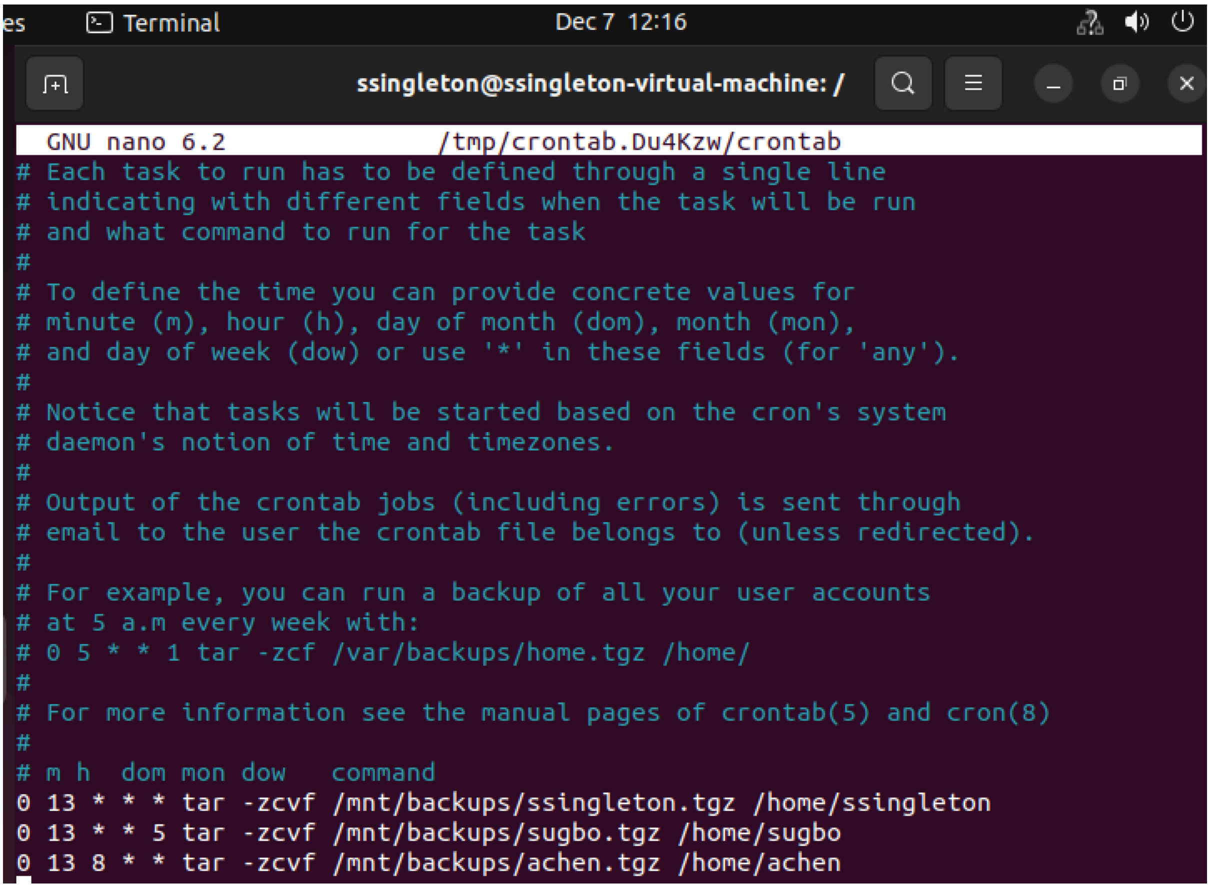
3. Backups

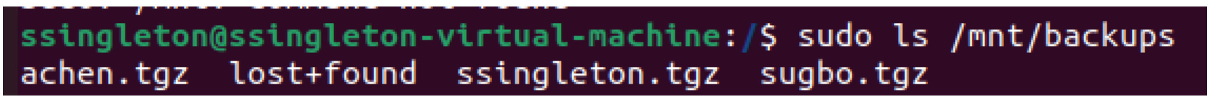
Steps to setup crontab nano

Cd /etc

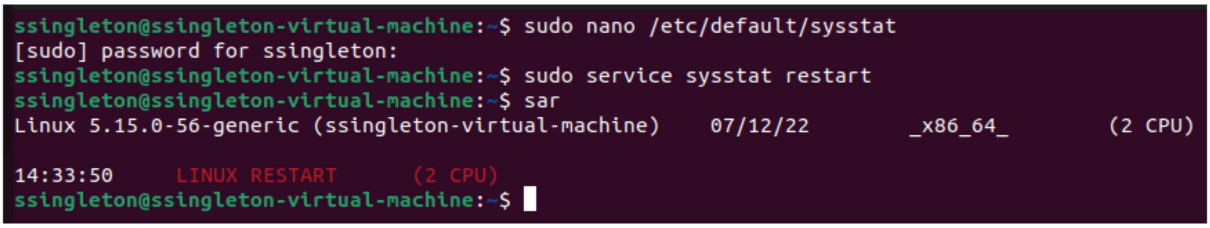
Crontab -e

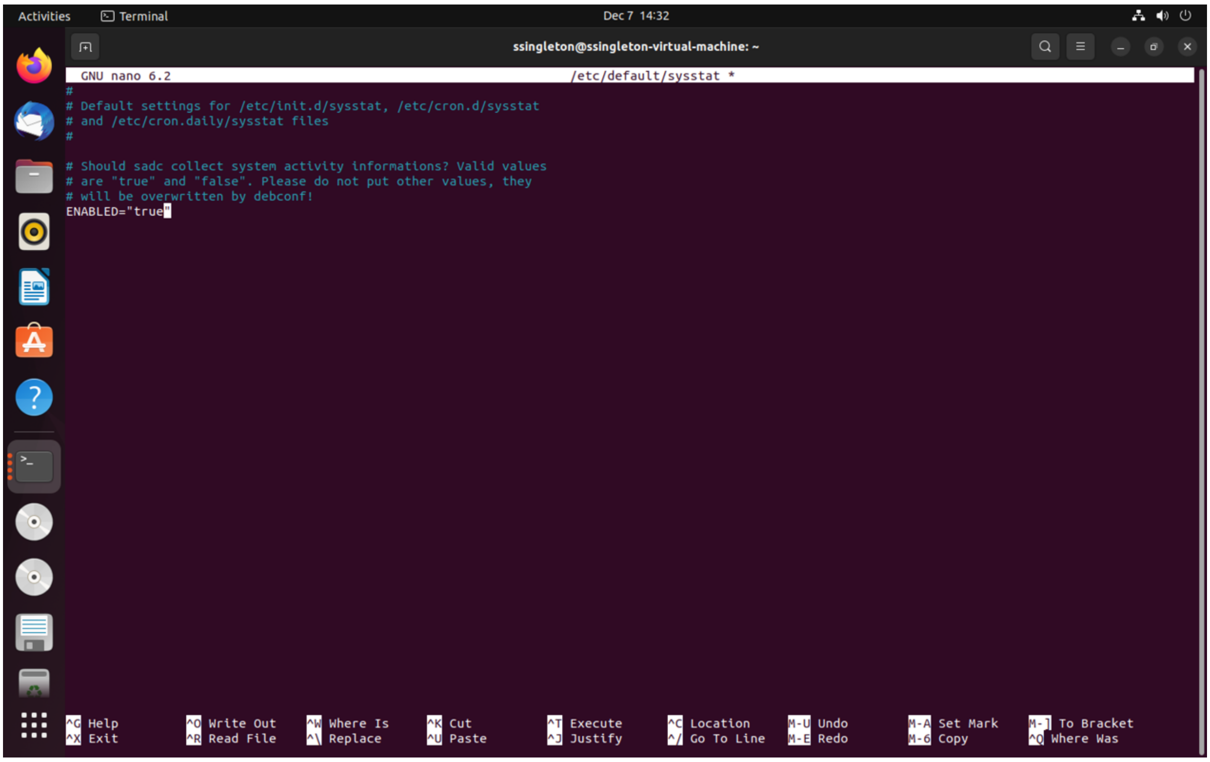
Chose 1. nano





4. Performance Monitoring





5.1. Script and logging rule

Nano to enter below

```
ssingleton@ssingleton-virtual-machine: ~
  GNU nano 6.2                        synflood.sh
#1/bin/bash
syn=$(cat /proc/sys/net/ipv4/tcp_syncookies)

if [ "$syn" -eq 1 ]; then
        echo "enabled"
        exit
else
        while true; do
                count=$(netstat -nt | grep SYN_RECV | wc -l)

                echo $count
                if [ $count -gt 30 ]; then

                        logger -t ATTACK "SYN FLOOD attack"
                        sleep 5m
                else
                        echo "everything okay"
                fi
        done
fi
                                [ Read 20 lines ]
^G Help       ^O Write Out ^W Where Is  ^K Cut        ^T Execute    ^C Location
^X Exit       ^R Read File ^\ Replace   ^U Paste      ^J Justify    ^/ Go To Line
```

```
ssingleton@ssingleton-virtual-machine:~$ cat /proc/sys/net/ipv4/tcp_syncookies
1
ssingleton@ssingleton-virtual-machine:~$ sudo bash synflood.sh
enabled
```

```
ssingleton@ssingleton-virtual-machine:~$ cd /etc/rsyslog.d
ssingleton@ssingleton-virtual-machine:/etc/rsyslog.d$ sudo nano 00-seans.conf
```

```
ssingleton@ssingleton-virtual-machine: /etc/rsyslog.d
  GNU nano 6.2                    00-seans.conf *
:syslogtag, isequal, "ATTACK:"         /var/log/synflood.log
```

5.2. Systemd Service

```
ssingleton@ssingleton-virtual-machine:/etc/rsyslog.d$ sudo nano seans.service
```

```
ssingleton@ssingleton-virtual-machine: /etc/rsyslog.d

  GNU nano 6.2                    seans.service *
[UNIT]
Description=

[SERVICE]
ExecStart=/home/ssingleton/synflood.sh

[INSTALL]
wantedBy=multi-user.target
```

6. Custom Linux Kernel

2.3

```
ssingleton@ssingleton-virtual-machine:~$ wget https://cdn.kernel.org/pub/linux/
kernel/v5.x/linux-5.9.12.tar.xz
--2022-12-07 16:29:26--  https://cdn.kernel.org/pub/linux/kernel/v5.x/linux-5.9
.12.tar.xz
Resolving cdn.kernel.org (cdn.kernel.org)... 199.232.25.176, 2a04:4e42:43::432
Connecting to cdn.kernel.org (cdn.kernel.org)|199.232.25.176|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 115553420 (110M) [application/x-xz]
Saving to: ‘linux-5.9.12.tar.xz’

linux-5.9.12.tar.xz 100%[===================>] 110.20M  19.9MB/s    in 6.6s

2022-12-07 16:29:34 (16.7 MB/s) - ‘linux-5.9.12.tar.xz’ saved [115553420/115553
420]
```

```
ssingleton@ssingleton-virtual-machine:~$ tar xJf linux-5.9.12.tar.xz
ssingleton@ssingleton-virtual-machine:~$ ls
Desktop    Downloads     linux-5.9.12.tar.xz  Pictures  snap         Templates
Documents  linux-5.9.12  Music                Public    synflood.sh  Videos
```

3.

```
gpg:                using RSA key 647F28654894E3BD457199BE38DBBDC86092693E
```

```
gpg: Signature made Wed 02 Dec 2020 07:55:33 GMT
gpg:                using RSA key 647F28654894E3BD457199BE38DBBDC86092693E
gpg: Can't check signature: No public key
```

Key ends with 6092693E and doesn’t say bad signature